Queen Mary
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EBU5376
Digital Signal Processing

Week 3 – Tutorial 2

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Question 1a

Use the DTFT to derive the frequency response for the system represented by the difference equation:

$$y[n] = 0.5x[n] + 0.5x[n-1]$$

Answer:

We obtain the system's impulse response $h[n]$ by putting $x[n] = \delta[n]$

Then apply the DTFT to h[n] to obtain H(ω):

$$H(\omega) = h[0]e^{-j\omega 0} + h[1]e^{-j\omega 1}$$

$$H(\omega) = 0.5 + 0.5e^{-j\omega}$$

Question 1b

Write an expression for each of the magnitude and phase of the frequency response of the system whose difference equation is:

$$y[n] = 0.5x[n] + 0.5x[n-1]$$

Then calculate the Magnitude and Phase for $\omega = 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi$ rads

Answer:

Magnitude of H(ω) =

$$|0.5 + 0.5e^{-j\omega}| = \sqrt{(0.5 + 0.5\cos\omega)^2 + (0.5\sin\omega)^2}$$

Phase of H(ω) =

$$\tan^{-1}\left(\frac{-0.5\sin\omega}{0.5 + 0.5\cos\omega}\right)$$

For $\omega = 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi$ rads

Magnitude = 1.000, 0.924, 0.710, 0.383, 0.000

Phase = 0°, -22.50^o, -45.00^o, -67.50^o, -90.00^o

(= 0 rads, -0.393 rads, -0.785 rads, -1.178 rads, -1.571 rads)

Question 2a

Use z-transforms to derive the frequency response for the system represented by the difference equation:

$$y[n] = x[n] - 0.5y[n-1]$$

Answer:

Taking z transforms of both sides gives $Y(z) = X(z) - 0.5z^{-1}$Y(z)

So the system's transfer function

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1}{1 + 0.5z^{-1}}$$

Putting z=$e^{j\omega}$ gives

$$H(\omega) = \frac{1}{1 + 0.5e^{-j\omega}}$$

Question 2b

Write an expression for each of the magnitude and phase of the frequency response of the system whose difference equation is:

$$y[n] = x[n] - 0.5y[n-1]$$

Then calculate the Magnitude and Phase for $\omega = 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi$ rads

Answer:

Magnitude of H(ω) = $\frac{1}{\sqrt{(1+0.5\cos\omega)^2+(0.5\sin\omega)^2}}$

Phase of H(ω) = $-\tan^{-1}(\frac{-0.5\sin\omega}{1+0.5\cos\omega})$

For ω = 0, π/4, π/2, 3π/4 and π rads

Magnitude = 0.670, 0.715, 0.894, 1.357, 2.000

Phase = 0°, 14.64^o, 26.57^o, 28.68^o, 0°

(= 0 rads, 0.26 rads, 0.46 rads, 0.5 rads, 0 rads)

Question 3

A system's "Characteristic Equation" is the denominator of its transfer function equated to 0.

Determine the stability of the sampled data systems whose characteristic equations are:

a) z^2-0.25=0

b) z^2-z+0.25=0

c) z^2-2z+1=0

d) $z^2 + 1.414$z+1=0

Question 3a

Answer:

z^2-0.25=0

Solve quadratic:

$$z = \frac{-0 \pm \sqrt{0 + 4 \times 0.25}}{2}$$

$$z = \pm 0.5$$

Poles lie inside the unit circle, therefore the system is stable

Question 3b

Answer:

z^2-z+0.25=0

Solve quadratic

$$z = \frac{+1 \pm \sqrt{1 - 4 \times 0.25}}{2}$$

$$z_1 = z_2 = 0.5$$

Poles lie inside the unit circle, therefore the system is stable

Question 3c

Answer:

z^2-2z+1=0

This equation factorises into real roots:

z^2-2z+1=(z-1)(z-1)

That is, there is a double pole at z=+1

These poles lie on the unit circle, so the system is Untable.

Question 3d

$z^2 + 1.414z+1=0$

Solve Quadratic

$$z = \frac{-1.414 \pm \sqrt{2-4}}{2}$$

$$z = \frac{-\sqrt{2} \pm j\sqrt{2}}{2} = \frac{-1}{\sqrt{2}} \pm \frac{j}{\sqrt{2}}$$

The magnitude of these from the origin is $(\frac{1}{2} + \frac{1}{2}) = 1$

So both poles lie on the unit circle and the system is marginally stable.

That is, the borderline between stable (converging output) and unstable (diverging output).

Question 4a

Find the transfer function of the system whose difference equation is:

$$y[n] = \frac{1}{3}\left(y[n-1] - y[n-2] + x[n-1] + x[n-2]\right)$$

Hence determine the system's stability.

Answer:

Re-write as: 3y[n] – y[n-1] + y[n-2] = x[n-1] +x[n-2]

Then taking Z transforms:

$3Y(z) - z^{-1}Y(z) + z^{-2}Y(z) = z^{-1}X(z) + z^{-2}X(z)$

Then $\frac{Y(z)}{X(z)} = \frac{z^{-1} + z^{-2}}{3 - z^{-1} + z^{-2}}$

Question 4b

Determine the stability of the system whose transfer function is:

$$H(z) = \frac{z^{-1} + z^{-2}}{3 - z^{-1} + z^{-2}}$$

Answer:

Re-write as: $\frac{z+1}{3z^2 - z + 1}$. Then find the system poles by putting the denominator = 0 (the system's "characteristic equation") and finding its roots.

$3z^2$ - z + 1 = 0; $z = \frac{1 \pm \sqrt{1 - 12}}{6} = \frac{1 \pm 3.32j}{6}$ = 0.17 $\pm 0.55j$

Magnitude of z = 0.58

Therefore the poles are inside the unit circle and the system is stable.

Zero Order Hold

When we introduced sampled systems and referred to digital-to-analog conversion (DAC), we introduced the concept of a Zero-Order Hold (ZOH). This circuit holds the value of a sample till the next sample is taken.

Question 5

Using Laplace Transforms, determine the Transfer Function of a ZOH.

Answer:

The Laplace transform of the unit step function $u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$

is given by

$$X(s) = \int_0^\infty u(t)e^{-st}\,dt = \int_0^\infty e^{-st}\,dt = \left.\frac{e^{-st}}{-s}\right|_{t=0}^{t=\infty} = \frac{1}{s}$$

Time delay property: Multiply the Laplace of a function by e^{-Ts}, T represents the sampling interval

So Transfer Function of a ZOH = $\frac{1}{s}$ - $\frac{1}{s}e^{-Ts}$ = $\frac{1-e^{-Ts}}{s}$,

Question 6

The transfer function of a ZOH is G(s) = $\frac{1-e^{-Ts}}{s}$

Determine the magnitude and phase of its frequency response function.

Answer:

G(ω) = $\frac{1-e^{-j\omega T}}{j\omega}$ = $\frac{1-(\cos\omega T - j\sin\omega T)}{j\omega}$

|G(ω)| = $\frac{\sqrt{(1-\cos\omega T)^2+(\sin\omega T)^2}}{\omega}$

Phase of G(ω) = $\{\tan^{-1}(\frac{\sin\omega T}{1-\cos\omega T})\}$ - 90°

Question 7

Given the pole-zero plot of a system:

- Zeros at $z = 1$ and $z = -1$
- Poles at $z = 0.9e^{-j\frac{\pi}{4}}$ and $z = 0.9e^{j\frac{\pi}{4}}$

Sketch the approximate amplitude response and explain the behavior.

Question 7

Zeros at $z = 1$ and $z = -1$:

1. $z = 1$ corresponds to $\omega = 0$ (DC)
2. $z = -1$ corresponds to $\omega = \pi$ (Nyquist frequency)
3. These lie on the unit circle

Zeros on the unit circle indicates that these frequencies are completely attenuated, i.e., the **amplitude response is zero at $\boldsymbol{\omega = 0}$ and $\boldsymbol{\omega = \pi}$**

Question 7

Poles at $z = 0.9e^{-j\frac{\pi}{4}}$ and $z = 0.9e^{j\frac{\pi}{4}}$:
These are inside the unit circle, hence the system is stable.
The pole is where the filter will amplify signals most strongly, and the angle of the pole represents the frequency that is amplified.

Hence the amplitude response plot has **a peak at $\pi/4$**

*The closer the poles are to the unit circle, the sharper the peak is.

Question 7

Zeros at $z = 1$ and $z = -1$

Poles at $z = 0.9e^{-j\frac{\pi}{4}}$ and $z = 0.9e^{j\frac{\pi}{4}}$:

We can also derive the **amplitude at the peak**, $|H(\omega = \pi/4)|$

First, write down the transfer function

$$H(\omega) = \frac{(z-1)(z+1)}{(z-0.9e^{-\frac{j\pi}{4}})(z-0.9e^{\frac{j\pi}{4}})} = \frac{z^2-1}{z^2 - 2\cdot 0.9\cos\frac{\pi}{4}\cdot z + 0.9^2} = \frac{z^2-1}{z^2 - 1.2728z + 0.81}$$

Then, compute the amplitude response

$$|H(\omega)| = \frac{\left|e^{j2\omega} - 1\right|}{\left|e^{j2\omega} - 1.2728e^{j\omega} + 0.81\right|}$$

Question 7

Given $\omega = \pi/4$, we have

$$e^{j2\omega} = e^{j\pi/2} = j$$

And

$$e^{j\omega} = e^{j\pi/4} = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}j$$

Therefore,

$$|H(\omega = \pi/4)| = \frac{|j-1|}{\left|j - 1.2728\left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}j\right) + 0.81\right|}$$

$$= \cdots = \frac{|j-1|}{|-0.09 + 0.1j|} = \frac{\sqrt{2}}{0.1346} \approx 10.51$$

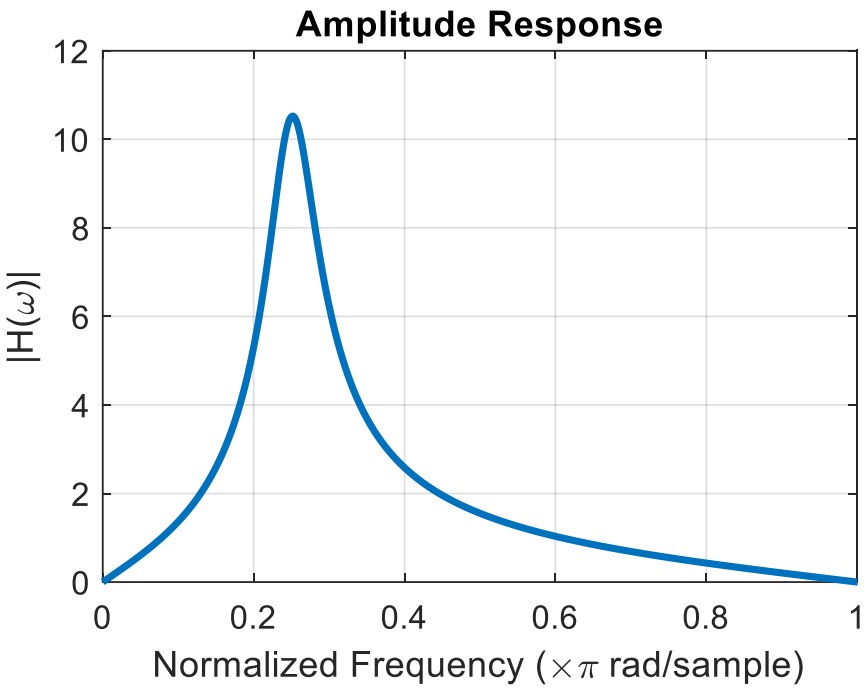
Question 7

Nulls at $\omega = 0$ and $\omega = \pi$ (due to zeros).

Peak near $\omega = \pi/4$ (due to poles).

Peak magnitude ≈ 10.51

The system amplifies frequencies around $\pi/4$, while attenuating DC, Nyquist, and other components.



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